YOLO Model Performance Evaluation Report

Blooming Stage Detection | Model Selection Analysis

Prepared: May 2026 | Task: Multi-class Object Detection | Classes: 5

1. Executive Summary

This report evaluates three YOLO model variants: YOLOv11 (One-stage), YOLOv8 (One-stage), and YOLOv2/6 (One-stage, referred to as YOLO26): benchmarked against a 12-image, 12-instance validation dataset encompassing five sugarcane growth stage classes: Bud Formation, Flowering, Mature Cane, Seedling, and Vegetative.

Based on holistic analysis across mAP50, mAP50-95, Precision, and Recall metrics, YOLOv8 is the recommended model for production deployment. It achieves the most balanced overall performance with an mAP50 of 0.441 and mAP50-95 of 0.260, and demonstrates the most reliable class-level consistency, particularly excelling in the critical Flowering class (mAP50: 0.913). YOLO26 shows competitive precision but fails on Recall for key classes. YOLOv11 underperforms significantly across all aggregate metrics and is not recommended for this task.

2. Evaluation Methodology

2.1 Dataset Overview

The validation set comprises 12 images and 12 annotated instances distributed across five classes. Class distribution is as follows:

Class	Images	Instances	Notes
Bud_Formation	2	2	Early stage; limited samples
Flowering	3	3	Critical phenological marker
Mature_Cane	2	2	Harvest readiness indicator
Seedling	4	4	Largest class in dataset
Vegetative	1	1	Smallest class; high variance risk

2.2 Evaluation Metrics

The following metrics are used for model assessment:

- **Precision (P):** The ratio of true positive detections to all positive predictions. High precision indicates fewer false alarms.
- **Recall (R):** The ratio of true positives to all ground truth instances. High recall indicates fewer missed detections.
- **mAP50:** Mean Average Precision at IoU threshold 0.50. The primary detection performance metric widely used in agricultural CV tasks.
- **mAP50-95:** Mean Average Precision averaged across IoU thresholds from 0.50 to 0.95 in 0.05 steps. Reflects localisation accuracy and robustness.

3. Model Performance Results

3.1 Overall (All-Class Aggregate) Metrics

Model	Precision	Recall	mAP50	mAP50-95
YOLOv11	0.036	0.900	0.177	0.087
YOLOv8	0.280	0.550	0.441	0.260
YOLO26	0.603	0.571	0.362	0.178

Table Legend: ***Green = high performance****,* ***Yellow = moderate****,* ***Red = low performance.***

3.2 Per-Class Detailed Metrics

YOLOv11

Class	Precision	Recall	mAP50	mAP50-95
Bud_Formation	0.002	1.000	0.133	0.093
Flowering	0.073	1.000	0.421	0.167
Mature_Cane	0.100	0.500	0.124	0.074
Seedling	0.003	1.000	0.084	0.031
Vegetative	0.001	1.000	0.124	0.071

YOLOv8

Class	Precision	Recall	mAP50	mAP50-95
Bud_Formation	0.000	0.000	0.031	0.019
Flowering	0.444	1.000	0.913	0.623
Mature_Cane	0.475	0.500	0.448	0.209
Seedling	0.322	0.250	0.315	0.103
Vegetative	0.158	1.000	0.497	0.348

YOLO26

Class	Precision	Recall	mAP50	mAP50-95
Bud_Formation	1.000	0.000	0.071	0.007
Flowering	0.714	0.854	0.746	0.540
Mature_Cane	0.232	1.000	0.745	0.239
Seedling	1.000	0.000	0.106	0.060
Vegetative	0.069	1.000	0.142	0.044

4. Comparative Analysis

4.1 YOLOv11 Poor Overall Performance

YOLOv11 records the lowest mAP50 (0.177) and mAP50-95 (0.087) of all three models evaluated. While it achieves a near-perfect Recall of 0.90 at the aggregate level, this is misleading: the corresponding Precision of 0.036 indicates the model is generating an extremely high volume of false positive detections. In practice, the model is detecting objects indiscriminately rather than learning discriminative features.

At the class level, four of five classes show Precision below 0.10, confirming that the model has not converged on meaningful class boundaries. The Flowering class is a relative exception (mAP50: 0.421), but this alone does not compensate for the model's failures across other classes. YOLOv11 is not suitable for the current task in its current training state.

4.2 YOLOv8 Best Balanced Performance (Recommended)

YOLOv8 achieves the highest mAP50 (0.441) and mAP50-95 (0.260) of all models, making it the top performer on both primary evaluation metrics. Its Precision (0.280) and Recall (0.550) are meaningfully balanced, suggesting the model avoids the extreme precision-recall trade-off issues seen in the other two models.

Key class-level highlights for YOLOv8:

- Flowering: Exceptional performance with mAP50 of 0.913 and mAP50-95 of 0.623 — the single best class result across all models in the evaluation.
- Vegetative: Strong mAP50 (0.497) and mAP50-95 (0.348), despite having only one validation image, showing reasonable generalisation.
- Mature_Cane: Solid mAP50 of 0.448 with the highest precision for this class (0.475) among all models.
- Seedling: Moderate mAP50 (0.315); room for improvement given it is the largest class in the dataset.
- Bud_Formation: The weakest class for YOLOv8, with 0.0 Precision and Recall, indicating the model fails to detect this stage. Additional training data for this class is strongly advised.

4.3 YOLO26 High Precision, Poor Recall Consistency

YOLO26 records the highest aggregate Precision (0.603) but this figure masks a critical structural problem: the model achieves zero Recall for both Bud_Formation and Seedling — two classes comprising 50% of the dataset instances. A Precision of 1.000 with Recall of 0.000 for these classes means the model is making no predictions at all, which artificially inflates the precision metric.

YOLO26 performs exceptionally on Flowering (mAP50: 0.746, mAP50-95: 0.540) and Mature_Cane (mAP50: 0.745), but its inability to reliably detect Bud_Formation and Seedling makes it unsuitable as a general-purpose deployment model for this task. It may be considered as a specialist model for a two-class or three-class subset application in future work.

5. Recommendation

Criterion	YOLOv11	YOLOv8 ✓	YOLO26
Overall mAP50	0.177 ✗	0.441 ✓	0.362
Overall mAP50-95	0.087 ✗	0.260 ✓	0.178
P/R Balance	Very Poor	Balanced ✓	Uneven

Criterion	YOLOv11	YOLOv8 ✓	YOLO26
Class Coverage	4/5 classes poor	4/5 classes good ✓	3/5 classes good
Best Class (mAP50)	Flowering: 0.421	Flowering: 0.913 ✓	Mature: 0.745
Production Ready	No ✗	Yes ✓	Partial

Recommendation: Deploy YOLOv8 for production use.

YOLOv8 is the only model that demonstrates acceptable performance across the majority of growth stage classes, with the highest overall mAP50 and mAP50-95. It provides a practical balance between detection confidence and coverage, and its architecture is well-supported with extensive community tooling for fine-tuning and deployment.

6. Future Improvement

6.1 Data Augmentation and Expansion

- Increase training samples for Bud_Formation and Vegetative classes, which are underrepresented (1 2 images each). A minimum of 30–50 images per class is recommended for reliable YOLO training.
- Apply targeted augmentation (colour jitter, rotation, crop-and-paste) specifically for Bud_Formation, where all three models underperform.
- Collect additional Seedling images to capitalise on YOLOv8's moderate existing performance for this class.

6.2 Model Fine-Tuning

- Fine-tune YOLOv8 with class-weighted loss to address the Bud_Formation failure (0.0 Precision and Recall). This is a critical gap for practical agricultural monitoring.
- Experiment with YOLOv8-medium or YOLOv8-large variants if compute resources allow, as larger backbone models may improve feature discrimination for morphologically similar growth stages.
- Consider ensemble strategies combining YOLOv8 (strong on Flowering, Vegetative) and YOLO26 (strong on Mature_Cane, Flowering) for a two-stage detection pipeline in high-stakes deployments.

6.3 Evaluation

- Re-evaluate on a larger held-out test set (minimum 50 images) to obtain statistically robust mAP estimates. The current 12-image validation set introduces high variance in class-level metrics, particularly for single-image classes.
- Report confidence interval bounds on mAP scores in future evaluations.

7. Conclusion

Among the three YOLO variants tested, **YOLOv8** is the clear choice for deployment in a sugarcane growth stage detection system. It achieves the best aggregate mAP50 (0.441) and mAP50-95 (0.260), demonstrates strong performance in the Flowering and Vegetative classes, and maintains a balanced precision-recall trade-off. YOLOv11 shows critical underfitting with near-zero precision, while YOLO26, despite high precision on some classes, fails to detect Bud_Formation and Seedling reliably.

Immediate next steps should focus on expanding the Bud_Formation training data and applying class-weighted fine-tuning to address the remaining gap in YOLOv8's performance profile. With targeted improvements, YOLOv8 is well-positioned to serve as the core detection backbone for precision agriculture applications in sugarcane cultivation monitoring.